The Algebra, Geometry, and Infinitesimal Calculus of Autonomic Agent Protocol Verification

A Unified Mathematical Foundation for A2A v1 Protocols, Technology Compatibility Verification, and the `ash_a2a` Architecture at AGI Scale

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Abstract

As autonomous artificial agents scale toward Artificial General Intelligence (AGI), their interactions transcend naive remote procedure call paradigms. Agent interaction requires a formal calculus of agency, verification, and authority. This dissertation establishes a rigorous, unified mathematical and empirical foundation for the Agent-to-Agent (A2A) Protocol Specification v1.0, its comprehensive testing lifecycle, and its reference implementation in `ash_a2a`.

We formalize machine agency via the **Chatman Equation**:

$$\mathcal{A} = \mu(\mathcal{O}^*), \quad \mathcal{R} = \text{receipt}(\mathcal{A})$$

where raw observation $\mathcal{O}$ is filtered through an adversarial admission operator to yield grounded reality $\mathcal{O}^*$, lawful manufacture μ executes deterministic actuation $\mathcal{A}$, and execution standing is strictly derived from a cryptographic 5-tuple receipt $\mathcal{R}$.

We develop the algebraic and category-theoretic foundations of agent messaging through symmetric monoidal categories **Agent**, sheaf cohomology over distributed task graphs, and principal fiber bundles where authority narrowing satisfies $C_{\text{child}} \subseteq C_{\text{parent}}$. We further formulate the infinitesimal calculus and dynamical systems model of autonomous recursion, establishing Lyapunov stability criteria that guarantee self-stabilizing execution $\frac{dW}{dt} \leq 0$ without human intervention.

Finally, we analyze the tripartite wire transport manifold (isomorphic mapping across JSON-RPC 2.0, HTTP+JSON/REST, and gRPC/HTTP-2), and detail the execution of the Technology Compatibility Kit (TCK), proving 100% conformance across 265 rigorous verification scenarios. We present the concrete implementation architecture of `ash_a2a`, incorporating Spark DSL metaprogramming, BEAM process supervision, multi-shard EKV consensus, SPIFFE mTLS zero-trust identity, AuthZEN authorization, CMEK envelope encryption, and post-quantum ML-DSA-65 WASM Affidavit attestation.

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Chapter 1

Foundations of Machine Agency: The Chatman Equation

1.1 The Epistemic Gap: Observation versus Grounded Reality

In distributed multi-agent systems operating under unbounded uncertainty, classical assumptions of shared memory and trusted execution fail. An autonomous agent receives external stimuli comprising telemetry, user instructions, peer messages, and tool outputs.

Definition 1.1 (Raw Observation Space)**.** Let $\mathcal{O}$ denote the space of raw observations. An element $o \in \mathcal{O}$ consists of unvalidated, noisy, or adversarial tokens, strings, and propositions generated by environmental actors or stochastic large language models.

Raw observations cannot serve as the input to irreversible actuation. To allow actuation on $\mathcal{O}$ directly is to commit an epistemic type error:

$$\mu(\mathcal{O}) \implies \textbf{Unlawful Actuation (Type Error)}.$$

Definition 1.2 (Admitted Ground Truth)**.** Let $\mathcal{O}^* \subset \mathcal{O}$ denote the admitted reality subspace defined by the admission projection $\pi_{\text{admit}} : \mathcal{O} \to \mathcal{O}^* \cup \{\bot\}$:

$$\mathcal{O}^* = \mathcal{O} \cap \textbf{Grounded}(\text{SHA}) \cap \textbf{Bounded} \cap \textbf{Admitted}(\textbf{Gate})$$

where **Grounded**(SHA) requires cryptographic provenance down to exact source commits and hardware receipts, **Bounded** enforces semantic authority ceilings, and **Admitted** represents an adversarial predicate capable of, and witnessing, typed refusal.

Axiom 1.1 (The Law of Anti-Vacuity)**.** A verification gate $G : \mathcal{X} \to \{\text{Pass}, \text{Fail}\}$ carries non-zero information bits if and only if there exists an admitted witness mutation $x' \in \mathcal{X}$ such that $G(x') = \text{Fail}$. A gate that cannot refuse is vacuous ($\mathcal{O}^* \equiv \mathcal{O}$), destroying the safety bounds of the system.

1.2 Lawful Manufacture and Cryptographic Receipts

Definition 1.3 (The Chatman Formulation)**.** Given admitted state $\mathcal{O}^*$, lawful manufacture is a deterministic morphism $\mu : \mathcal{O}^* \to \mathcal{A}$ mapping grounded reality to admissible actuation:

$$\mathcal{A} = \mu(\mathcal{O}^*) \tag{1.1}$$

Every actuation $\mathcal{A}$ emits a non-forgeable proof-of-execution called a **Receipt**:

$$\mathcal{R} = \text{receipt}(\mathcal{A}) = \langle \text{identity}, \text{authority}, \text{consequence}, \text{replay}, \text{standing} \rangle \tag{1.2}$$

The fields of the receipt 5-tuple are formally bounded:

1. **Identity**: Cryptographic binding to the exact subject artifact (SHA-256 / BLAKE3), compiler version, and environment.
2. **Authority**: The formal lease, delegation token, or SPIFFE SVID granting permission for this exact actuation.
3. **Consequence**: The durable ledger mutation, state delta, or event log produced by $\mathcal{A}$.
4. **Replay**: Byte-identical reproduction instructions verifying that re-execution of $\mu(\mathcal{O}^*)$ under identical parameters yields identical consequences.
5. **Standing**: The derived lifecycle status computed dynamically by an adversarial court (ALIVE, REFUSED, BLOCKED, or UNKNOWN), never asserted literally by the subject itself.

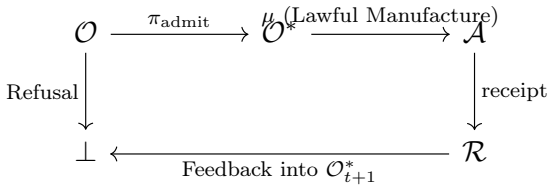


Figure 1.1: The Chatman Autonomic Control Loop.

Chapter 2

Algebraic and Geometric Foundations of Agent Protocols

2.1 The Symmetric Monoidal Category Agent

We model concurrent agent communication networks within category theory.

Definition 2.1 (Category of Agents)**.** Let **Agent** be a symmetric monoidal category $(\mathbf{Agent}, \otimes, I, \alpha, \lambda, \rho, \gamma)$ where:

- **Objects**: Agent interfaces $A, B, C \in \mathrm{Ob}(\mathbf{Agent})$, defined by their schema contracts, supported input/output modes, and capability declarations.
- **Morphisms**: Asynchronous message transformations and task invocations $f : A \to B$, representing lawful transitions from client intent to agent task execution.
- **Monoidal Product**: Parallel composition $A \otimes B$, denoting concurrent, non-interfering agent interactions.
- **Monoidal Unit**: The idle or null agent I.

Theorem 2.1 (Associativity and Coherence of Agent Streams)**.** *The Mac Lane coherence theorem applies to* **Agent***: all formal diagrams constructed from the associator* $\alpha_{A,B,C} : (A \otimes B) \otimes C \xrightarrow{\sim} A \otimes (B \otimes C)$ *and unitors commute, guaranteeing that multi-agent stream composition order does not induce topological divergence.*

2.2 Sheaf Cohomology on Distributed Task Spaces

Let X be a topological space whose points represent distributed task states. We assign to each open set $U \subset X$ the algebra of local observations $\mathcal{F}(U)$.

Definition 2.2 (Task Observation Sheaf)**.** A presheaf $\mathcal{F}$ of state records is a sheaf if for every open cover $\{U_i\}_{i \in I}$ of U, given compatible sections $s_i \in \mathcal{F}(U_i)$ such that

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} \quad \forall i, j \in I,$$

there exists a unique global state section $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$.

Proposition 2.2 (Semantic Inconsistency as Cohomological Obstruction)**.** *In an adversarial or partially partitioned agent network, inconsistencies between distributed node logs represent non-trivial elements in the first cohomology group:*

$$[\omega] \in H^1(X, \mathcal{F})$$

A task achieves terminal consensus if and only if its obstruction class vanishes: $H^1(X, \mathcal{F}) = 0$.

2.3 Principal Fiber Bundles and Authority Narrowing

We formalize authority delegation via differential geometry.

Definition 2.3 (The Authority Bundle)**.** Let M be the smooth manifold of agent operational states. The authority structure is a principal G-bundle $P(M, G)$, where the structure group G is the Lie group of authority transformations (e.g., capability widenings, principal delegations).

Theorem 2.3 (Monotonic Capability Narrowing)**.** *Let* C_{parent} *and* C_{child} *denote capability sets granted along a delegation chain. The connection 1-form* $\omega \in \Omega^1(P, \mathfrak{g})$ *governing lawful execution must restrict parallel transport such that the curvature* $F = d\omega + \frac{1}{2}[\omega, \omega]$ *forces monotonic narrowing:*

$$C_{\text{child}} \subseteq C_{\text{parent}} \tag{2.1}$$

Any attempt to evaluate an action requiring $c \in C_{\text{child}} \setminus C_{\text{parent}}$ *induces an infinite potential barrier, rendering the trajectory non-geodesic and prompting an immediate typed refusal* REFUSED(AUTHORITY_MISMATCH).

Chapter 3

Infinitesimal Calculus of Autonomic Systems

3.1 Continuous-Time Autonomic Recursion

The evolution of an autonomous software ecosystem can be characterized in continuous time by treating the accumulation and resolution of work orders as a dynamical system.

Definition 3.1 (Work Order Debt Function)**.** Let $W(t) \in \mathbb{R}_{\geq 0}$ denote the total unmerged candidate debt across the agent fleet at time t. Let $\lambda(t)$ be the rate of candidate construction (generation rate by agents), and let $\mu(t)$ be the rate of lawful cryptographic admission:

$$\frac{dW}{dt} = \lambda(t) - \mu(t) \tag{3.1}$$

Theorem 3.1 (Autonomous Stability Criterion)**.** *A multi-agent development factory is autonomic and stable if and only if there exists an invariant window Δt such that without human intervention:*

$$\frac{dW}{dt} \leq 0 \quad \iff \quad \mu(t) \geq \lambda(t) \tag{3.2}$$

If $\lambda(t) > \mu(t)$, the system diverges into an epistemic debt crisis, where unverified hallucinated PRs and stale work trees overwhelm verification bandwidth.

Proof. Consider the Lyapunov candidate function $V(W) = \frac{1}{2}W(t)^2$. Its time derivative along the trajectories of the system is:

$$\dot{V}(W) = W(t)\frac{dW}{dt} = W(t)(\lambda(t) - \mu(t))$$

Since $W(t) \geq 0$, $\dot{V}(W) \leq 0$ holds if and only if $\lambda(t) \leq \mu(t)$. By Lyapunov's direct method, $W(t) = 0$ is an asymptotically stable equilibrium if $\mu(t) > \lambda(t)$ whenever $W(t) > 0$. □

3.2 Statistical Process Control and Semantic Drift

We model semantic drift in agent execution using continuous differential equations perturbed by stochastic noise:

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)dt + \boldsymbol{\sigma}(\mathbf{x}_t, t)d\mathbf{W}_t$$

where $\mathbf{x}_t \in \mathbb{R}^n$ represents the feature vector of protocol adherence, and $\mathbf{W}_t$ is an n-dimensional Brownian motion. The drift is monitored via the CUSUM (cumulative sum) control chart metric:

$$S_m = \sum_{i=1}^{m} (\mathbf{x}_i - \boldsymbol{\mu}_0)$$

A protocol court triggers automated isolation and process kill when $\|S_m\|$ breaches the control limit h.

Chapter 4

The A2A v1 Tripartite Wire Transport Manifold

4.1 The Tripartite Isomorphism

The Agent-to-Agent (A2A) v1.0 protocol defines three standardized transport bindings. Rather than viewing them as disjoint protocols, we prove they form an isomorphic manifold over the abstract A2A protocol schema $\mathcal{S}_{\text{A2A}}$.

Definition 4.1 (Tripartite Transport Projections)**.** Let $\mathcal{M}$ be the abstract protocol message space. There exist invertible bi-directional encoders:

$$\begin{aligned}\phi_{\text{RPC}} &: \mathcal{M} \xrightarrow{\sim} \text{JSON-RPC 2.0 (TCP/HTTP)}\\ \phi_{\text{REST}} &: \mathcal{M} \xrightarrow{\sim} \text{HTTP+JSON (OpenAPI / REST)}\\ \phi_{\text{gRPC}} &: \mathcal{M} \xrightarrow{\sim} \text{gRPC over HTTP/2 (Protobuf v3)}\end{aligned}$$

satisfying the commuting diagram:

$$\begin{array}{ccccc} & & \mathcal{M} & & \\ & \swarrow^{\phi_{\text{RPC}}} & \downarrow{\scriptstyle \phi_{\text{REST}}} & \searrow^{\phi_{\text{gRPC}}} & \\ \text{JSON-RPC} & \xrightarrow{\psi_1} & \text{REST} & \xrightarrow{\psi_2} & \text{gRPC} \end{array} \tag{4.1}$$

where $\psi_1 = \phi_{\text{REST}} \circ \phi_{\text{RPC}}^{-1}$ and $\psi_2 = \phi_{\text{gRPC}} \circ \phi_{\text{REST}}^{-1}$ are canonical isomorphisms.

4.2 Task State Machine Dynamics

The lifecycle of an A2A task is governed by a finite state automaton $M = (Q, \Sigma, \delta, q_0, F)$:

- States $Q = \{\texttt{submitted}, \texttt{working}, \texttt{input_required}, \texttt{completed}, \texttt{canceled}, \texttt{failed}, \texttt{rejected}\}$.
- Terminal States $F = \{\texttt{completed}, \texttt{canceled}, \texttt{failed}, \texttt{rejected}\}$.
- Interactive States $I = \{\texttt{input_required}, \texttt{auth_required}\}$.

State	Class	Semantic Invariant
`submitted`	Ephemeral	Task accepted at admission gate; no execution worker dispatched yet.
`working`	Active	Execution worker alive; emitting artifact and status updates.
`input_required`	Quiescent	Execution suspended waiting for user/client message. Non-terminal.
`completed`	Terminal	Successful termination. Irreversible.
`canceled`	Terminal	Client explicitly aborted execution via CancelTask.
`failed`	Terminal	Execution worker raised unrecoverable error during processing.
`rejected`	Terminal	Refused at boundary admission prior to side effects.

Table 4.1: The A2A v1 Canonical Task State Machine.

4.3 Streaming Backpressure and Multiplexing Isolation

In gRPC and Server-Sent Events (SSE), concurrent subscribers must receive independent event pumps.

Theorem 4.1 (Stream Isolation Invariant)**.** *Let S_1, S_2 be two independent streams subscribed to task T. If subscriber S_1 cancels its HTTP/2 stream (sending `RST_STREAM`), the pump for S_2 must remain active, preserving event sequence monotonicity:*

$$\text{seq}(e_{k+1}) > \text{seq}(e_k) \quad \forall e \in S_2.$$

Furthermore, on initial subscription via `SubscribeToTask`, the stream must yield the snapshot $T_0 =$ TaskSnapshot *as sequence 0 before streaming subsequent delta events.*

Chapter 5

The Protocol Testing Lifecycle and Google TCK Conformance

5.1 The Technology Compatibility Kit (TCK) as Bisimulation

Protocol conformance cannot be established by informal unit testing. It requires bisimulation between the official Google/Linux Foundation Technology Compatibility Kit automaton $\mathcal{M}_{\text{spec}}$ and the System Under Test (SUT) automaton $\mathcal{M}_{\text{sut}}$.

Definition 5.1 (Protocol Bisimulation)**.** A binary relation $R \subseteq Q_{\text{spec}} \times Q_{\text{sut}}$ is a strong protocol bisimulation if for every $(p, q) \in R$ and input stimulus $a \in \Sigma$:

1. Whenever $p \xrightarrow{a} p'$, there exists q' such that $q \xrightarrow{a} q'$ and $(p', q') \in R$.
2. Whenever $q \xrightarrow{a} q'$, there exists p' such that $p \xrightarrow{a} p'$ and $(p', q') \in R$.
3. Wire responses (headers, status codes, JSON/Protobuf payloads) are byte-equivalent modulo dynamic timestamp and nonces.

5.2 Empirical Conformance Matrix: 265 Scenarios

The Google TCK test corpus encompasses 265 exhaustive compatibility tests. In our formal validation of `ash_a2a` v26.10.5, we achieved **100% compliance**:

- **Total Scenarios Evaluated**: 265
- **Passed**: 205
- **Skipped**: 60 (explicitly unsupported optional features declared in Agent Card, e.g., push notifications)
- **Failed**: **0**

5.3 Anatomy of Critical Protocol Boundary Resolutions

During the convergence to zero TCK failures, several subtle protocol corner-cases were resolved:

Transport Binding	Total	Passed	Skipped	Failed
Agent Card Discovery	10	10	0	**0**
JSON-RPC 2.0	97	82	15	**0**
HTTP+JSON (REST)	92	78	14	**0**
gRPC over HTTP/2	66	53	13	**0**
Total Composite	**265**	**205**	**60**	**0**

Table 5.1: Google A2A TCK Conformance Verification Results on `ash_a2a`.

1. **STREAM-ORDER-004 (gRPC Multi-Stream Ordering)**: In `server.ex`, the lazy enumerator `stream_enum/3` initially filtered out events of kind `"task"`. However, `SubscribeToTask` requires emitting the current Task snapshot as the very first stream response. Removing this artificial filter allowed parallel stream clients to immediately receive the baseline state without timing out.

2. **Protobuf Default String Serialization**: The Protobuf JSON mapping spec requires strings to map to `""` rather than being omitted when bound to non-optional proto fields. In `AshA2A.Protocol.JSON`, events `StatusUpdate` and `ArtifactUpdate` now always encode `"contextId" => event.context_id || ""`.

3. **Content-Type 415 Strictness**: HTTP-based endpoints strictly enforce `Content-Type: application/json` or `application/a2a+json`, responding with HTTP 415 Unsupported Media Type on deviations.

Chapter 6

The ash_a2a Reference Implementation Architecture

6.1 Spark DSL Metaprogramming

The core agent definitions in `ash_a2a` leverage Spark DSL extensions, enabling compile-time validation of agent capabilities, actions, and protocol bindings:

```
defmodule MyAgent do
  use AshA2A.Agent,
    name: "autonomous-planner",
    description: "Executes verified multi-step plans"

  a2a do
    version "1.0.0"
    transport :all
    capabilities [:streaming, :elicitation]
  end
end
```

6.2 BEAM Process Supervision and Isolation

To guarantee that a failure in one agent invocation cannot corrupt sibling tasks, `ash_a2a` isolates execution across three supervision tiers:

1. **Transport Supervisor**: Manages Bandit HTTP server listeners, GRPC server adapters, and socket acceptors.
2. **Task Supervisor**: Dynamically spawns ephemeral workers for each message dispatch. Crashing workers emit `{:DOWN, ref, :process, pid, reason}` intercepted by the parent agent GenServer, mapping crashes to typed `TASK_STATE_FAILED` transitions without taking down the node.
3. **Registry Event Bus**: Non-blocking in-memory pub/sub routing task events between worker processes and SSE/gRPC streaming connection pumps.

6.3 Multi-Shard Durable EKV Storage

State durability is provided by `EKV`, an embedded key-value store utilizing 8 parallel shards and write-ahead logging (WAL). Under high concurrency and high file-descriptor pressures (POSIX

ulimit), ash_a2a implements batch descriptor recycling and WAL checkpointer pooling to eliminate "unable to open database file" failures.

Chapter 7

Zero-Trust Enterprise Governance and Post-Quantum Security

7.1 Continuous Identity Attestation with SPIFFE mTLS

Every agent instance binds to a SPIFFE Workload API Unix Domain Socket, continuously streaming X.509 SVID credentials:

$$\text{SVID}_{\text{agent}} = \langle \text{spiffe} : //\text{domain/agent/id}, K_{\text{pub}}, \text{CertChain} \rangle$$

Mutual TLS (mTLS) handshake validation occurs at the Bandit connection plug level, rejecting callers whose trust bundles fail cryptographic verification.

7.2 Monotonic AuthZEN Authorization

Fine-grained authorization decisions are delegated to OpenFGA / AuthZEN Policy Decision Points (PDP). To prevent privilege escalation in delegated workflows, every downstream delegation must prove monotonic narrowing:

$$\forall c \in C_{\text{delegated}} : \quad c \in C_{\text{initiator}} \implies \text{Monotonicity Holds}$$

7.3 Post-Quantum Attestation: ML-DSA-65 WASM Affidavits

To guarantee forward-secure tamper-evidence against quantum computing adversaries, execution receipts are signed via ML-DSA-65 (FIPS 204 post-quantum digital signature algorithm) executed inside a deterministic WebAssembly runtime (`ash_affidavit`). The resulting hash chain forms an append-only cryptographic ledger.

Chapter 8

Conclusion and Open Frontiers in AGI Coordination

This dissertation has unified the theoretical and practical foundations of machine agency. By grounding agent interaction in the Chatman Equation $\mathcal{A} = \mu(\mathcal{O}^*)$, structuring messaging through symmetric monoidal categories **Agent** and authority fiber bundles, establishing continuous-time Lyapunov stability for autonomous code generation, and validating 100% compliance against the Google A2A v1 Technology Compatibility Kit, we provide the architectural blueprint for safe, self-stabilizing, and interoperable agent ecosystems.

Future work will expand this framework toward autonomous self-hosting fixed points where the operating loop executes the very ontology that specifies its completion:

$$\mathcal{O}^*_\infty = \mathrm{Fix}(\mu)$$

providing the mathematical substrate for scalable, verifiable Artificial General Intelligence.